

Submission Summary

Conference Name

4th International Conference on Networks, Multimedia, and Information Technology
(NMITCON-2026)

Paper ID

492

Paper Title

Artificial Intelligence in Cancer Therapy: An Enhanced Machine Learning Framework for Personalized Lung Cancer Treatment Prediction

Abstract

The difficulty encountered when trying to choose the best treatment for cancer is still an issue in the field of oncology today. Treatments are often prescribed based solely on what the doctor has seen work before, which results in delays, inconsistencies, and too much reliance on data from outside of the specific patient setting. The purpose of this study is to develop an artificial intelligence-based framework called OncoFusionNet-LC that can assist in identifying therapy options for patients suffering from lung cancer. The large synthetic lung cancer database (Bangladesh Perspective) was collected to build a comprehensive dataset of records (5,000 total) across 17 demographic, clinical and environmental characteristics. Four different machine learning algorithms including Logistic Regression, Decision Trees, Random Forest and SVMs were compared. The proposed OncoFusionNet-LC outperformed the other three algorithms achieving accuracy of 69.2% and AUC of 0.73. These results show that while there are a lot of participants with many confounding variables, artificial intelligence can be used as a reliable second opinion to help the oncologist provide better quality data driven personalized medicine.

Index Terms—Artificial Intelligence; Lung Cancer; Therapy Selection; Machine Learning; OncoFusionNet-LC; Personalized Oncology; Clinical Decision Support

Created

2026-04-08 16:03:16 GMT+5:30

Last Modified

2026-04-08 16:03:16 GMT+5:30

Authors

Vishwajeet Singh (JAIN UNIVERSITY) <vickeysingh596.25@gmail.com>

SYED FAIZAN PASHA (JAIN UNIVERSITY) <syedfaizanpasha16@gmail.com>

SUVENDU MATA (JAIN UNIVERSITY) <Suvendumata@gmail.com>

Submission Files

Research Paper - AI in Cancer Therapy.pdf (799.4 Kb, 2026-04-08 16:03:03 GMT+5:30)
